SIMATS ENGINEERING

ASSESSMENT TOOL 1: ANALYTICAL PROBLEM SOLVING

COURSE NAME: Cloud Computing and Big Data Analytics
COURSE CODE: CSA15**15**
COURSE OUTCOME COVERED: CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)
Assessment Tool Weightage: 40%
Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5
Total Marks: 50

Code of Conduct

I ,**P. Nitheesh Kumar(192524051)**certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: .**Nitheesh Kumar P**

Assessment Tasks – Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach

Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded: _____

ANSWERS

Answer 1: Type-1 vs Type-2 Hypervisors

Introduction: A hypervisor, also called a virtual machine monitor (VMM), creates and manages virtual machines (VMs) by allocating physical hardware resources to them. Type-1 hypervisors run directly on server hardware, while Type-2 hypervisors run as applications on top of a host operating system.

Performance: Type-1 hypervisors have a shorter path to the hardware and generally provide lower overhead, more predictable performance, and better resource control. This is important for mission-critical cloud applications with demanding CPU, memory, storage, and network requirements. Type-2 hypervisors depend on the host operating system, adding another software layer and potential overhead.

Security: Type-1 hypervisors reduce dependence on a general-purpose host OS, which can reduce the attack surface. They still require secure configuration, patching, access controls, and monitoring. Type-2 hypervisors inherit risks from both the host OS and the hypervisor application; compromise of the host OS can affect the VMs.

Manageability: Type-1 platforms commonly provide centralized management, live migration, high availability, resource scheduling, and monitoring suited to large datacenters. Type-2 products are convenient for desktop testing and development, but are usually less suitable for managing a large production fleet.

Conclusion: A Type-1 hypervisor is the better choice for a cloud datacenter hosting mission-critical applications because it generally offers stronger performance predictability, a smaller dependency chain, and datacenter-grade management. The provider should still apply regular security updates, least-privilege administration, network segmentation, backups, and continuous monitoring.

Answer 2: Workload Isolation and Attack Surfaces

Virtualization-based isolation: The provider can run finance, healthcare, and student-portal services in separate VMs or hardened containers, with separate virtual networks, access policies, storage permissions, and security groups. The hypervisor schedules CPU and memory while maintaining VM boundaries. Virtual switches and VLANs or software-defined network policies restrict traffic between workloads. Separate encryption keys and identity roles help prevent unauthorized access to data.

Attack surface 1 – Hypervisor and management plane: A hypervisor vulnerability or a compromised management console may allow an attacker to escape a VM boundary, control other VMs, or alter resource assignments. Reduce this risk by patching promptly, limiting management access to a protected network, enforcing multifactor authentication and role-based access control, and auditing administrative actions.

Attack surface 2 – Virtual networking and VM interfaces: Misconfigured virtual switches, overly permissive security groups, exposed VM services, or vulnerable virtual network components can allow lateral movement between tenants. Use default-deny network rules, micro-segmentation, firewalls, secure configuration baselines, vulnerability scanning, and continuous traffic monitoring.

Conclusion: Virtualization provides useful logical separation, but isolation is only as strong as the hypervisor, management plane, network configuration, and operational controls. Sensitive finance and healthcare workloads may also require dedicated keys, stricter access policies, and compliance-focused monitoring.

Answer 3: VM Host Utilization and Corrective Actions

Interpretation of metrics: CPU utilization of 92% indicates the host is close to saturation and may have insufficient compute capacity for peak demand. Memory utilization of 88% leaves a limited safety margin; memory pressure can cause ballooning, swapping, or VM performance degradation. Storage I/O latency of 35 ms is high for many interactive and transactional workloads and suggests storage contention, slow disks, queue saturation, or an overloaded storage network. An average VM boot time of 170 seconds may reflect resource contention, slow storage reads, lengthy startup services, or inefficient boot workflows.

Likely bottlenecks: The host appears constrained in both compute and memory, while storage latency is a particularly likely contributor to slow VM booting. These symptoms should be verified with time-series metrics rather than assuming one cause: inspect CPU ready/steal time, memory ballooning and swap activity, IOPS and queue depth, throughput, storage-network errors, and boot logs.

Corrective actions: 1. Identify high-consumption VMs and right-size their vCPU and memory allocations. 2. Migrate or rebalance VMs to less-loaded hosts; add host capacity if high utilization persists. 3. Check for memory overcommit, ballooning, and swapping; reserve memory for critical workloads where appropriate. 4. Investigate storage latency, queue depth, IOPS limits, and network congestion; move critical VMs to faster storage or distribute I/O-heavy workloads. 5. Use capacity monitoring and alerts, and schedule non-urgent batch jobs outside peak periods. 6. Review VM boot services and use approved templates or automation to reduce unnecessary startup work.

Conclusion: The combination of high CPU and memory use with 35 ms storage latency indicates resource contention. Prioritize diagnosing storage performance and host saturation, then rebalance workloads or scale resources. Measure boot time and application response time after each change to confirm improvement.

Answer 4: Software Defined Datacenter (SDDC) and Agility

Definition: A Software Defined Datacenter (SDDC) uses software to abstract, pool, and automate compute, storage, and networking resources. Instead of configuring every device manually, administrators define policies and use centralized software management to provision and control infrastructure.

Compute virtualization: A hypervisor pools server resources and runs multiple VMs on shared physical servers. Templates, orchestration, automated placement, and live migration can provision workloads quickly and help balance resource use. In a traditional datacenter, adding or configuring servers often requires more manual hardware and operating-system work.

Storage virtualization: Storage virtualization combines capacity from multiple physical devices into logical pools or datastores. Software-defined storage can automate provisioning, replication, snapshots, and policy-based placement. This improves flexibility and recovery, whereas traditional storage provisioning may require device-by-device configuration.

Network virtualization: Software-defined networking (SDN) separates network control from forwarding and enables centrally managed virtual networks, overlays, and policy-based segmentation. Networks can be created or changed through software and APIs rather than manually reconfiguring many switches and routers. Security policies can follow workloads as they move.

Agility and considerations: SDDC supports faster deployment, repeatable automation, better resource utilization, and more consistent policy enforcement. However, it requires skilled administration, reliable management software, careful access control, interoperability planning, and monitoring. A failure or compromise of the central management plane can have broad impact, so it should be highly available and strongly secured.

Conclusion: Compared with a traditional datacenter, an SDDC makes compute, storage, and networking more programmable and policy-driven. This improves agility and scalability when supported by sound governance, automation testing, and security controls.

Answer 5: Secure Identity and Privacy in Multi-Cloud Federation

Why identity mismatches occur: Cloud providers may use different identity formats, user identifiers, role names, authentication methods, token formats, and trust policies. A user who is an administrator in one provider may be mapped incorrectly in another. Differences in attribute names, group membership, account lifecycle, and token expiry can cause failed logins or excessive privileges.

Federated identity design: Adopt a trusted identity provider (IdP) or identity broker that supports standard federation protocols such as OpenID Connect (OIDC) or SAML 2.0, depending on provider support. Configure each cloud as a relying party/service provider and establish explicit trust relationships. Use consistent, immutable subject identifiers and documented attribute mappings; do not rely only on display names or email addresses for identity matching.

Authorization and security: Use single sign-on with multifactor authentication, least-privilege role mapping, and short-lived, audience-restricted tokens. Map central groups to narrowly scoped cloud roles, and deny access by default if required claims are missing or invalid. Validate token signatures, issuer, audience, expiry, and intended use. Use separate privileged accounts, just-in-time elevation where available, and strong audit logging across providers.

Privacy protections: Share only the identity attributes required for the requested service. Use pairwise or provider-specific pseudonymous identifiers where feasible to reduce cross-service tracking. Encrypt data in transit, protect signing keys in a managed key store or hardware security module, set retention limits for identity logs, and restrict who can view those logs. Assess data residency and applicable privacy requirements before moving personal or healthcare information across clouds.

Operations and recovery: Maintain a clear process for onboarding and offboarding users, rotating keys, reviewing trust relationships, revoking sessions, and responding to compromised identities. Test federation with representative user roles and negative cases, monitor authentication failures, and periodically audit cross-cloud permissions.

Conclusion: A secure multi-cloud federation should standardize protocols and identity mapping while keeping authorization explicit in each cloud. Strong authentication, minimal attribute sharing, short-lived tokens, careful key management, and consistent auditing resolve identity mismatches while protecting privacy.